Off-Road Illini BAJA SAE

CREO Spring Workshop

02/15/2026

Attendance Question: What does your dream Valentine's day look like ⁉️

Baja Context

1. **Summer design cycle**
   a. Prints 4 manufacturing later
2. **Fall design review**
   a. In references doc!!
3. **Creo → machining companies / trained machinists → car!**

Setup

1. **EWS connection 🤞**
    a. Open File Explorer.
    b. Browse to the U: drive.
        i. Make a folder!!!!!!!!!!!!!!
        ii. Make a subfolder (baja spring workshop, etc)
    c. Do ts on your pc side (later)
        i. Use EWS link in references sheet (also in zip!)
        ii. Save to another location before semester ends
2. **Open Creo! Search Creo Parametric 11.0.1.0 and click**

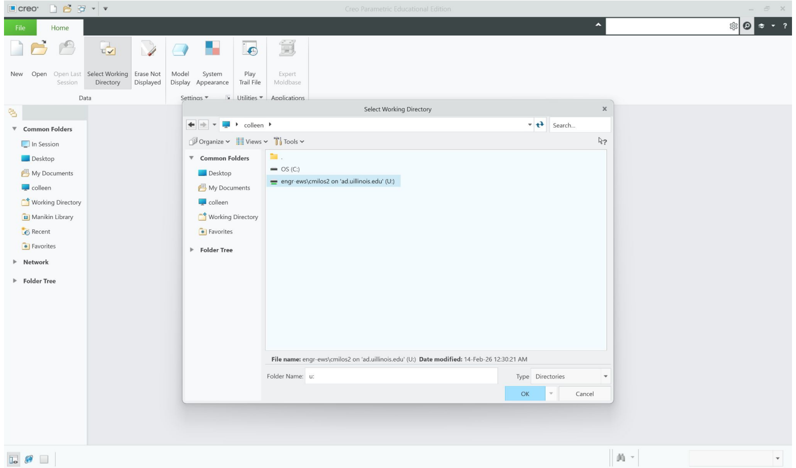
Check out/Open/Save

1. Check out the model by sending a message in slack #model2X.
2. Download the zip file from the main folder. (check your email!!)
3. Extract file, set as your **working directory**, and make all necessary edits.

U:\Baja\Spring workshop!!!\

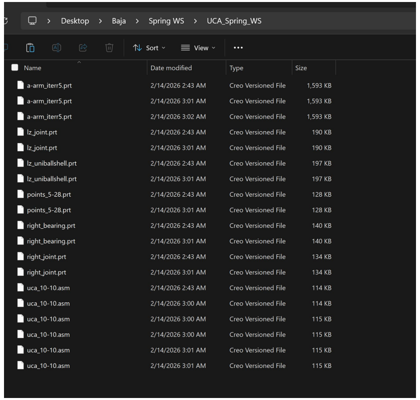
Put UCA zip
file after ts

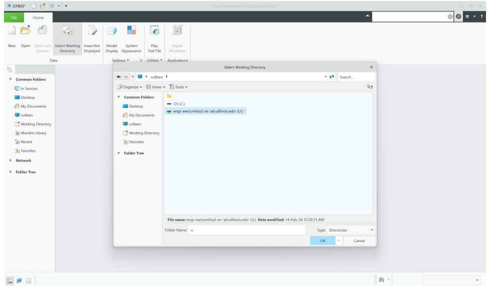
- Creo has weird saving tendencies
- **Working directories** are REALLY important for cloud integration



Working Directories

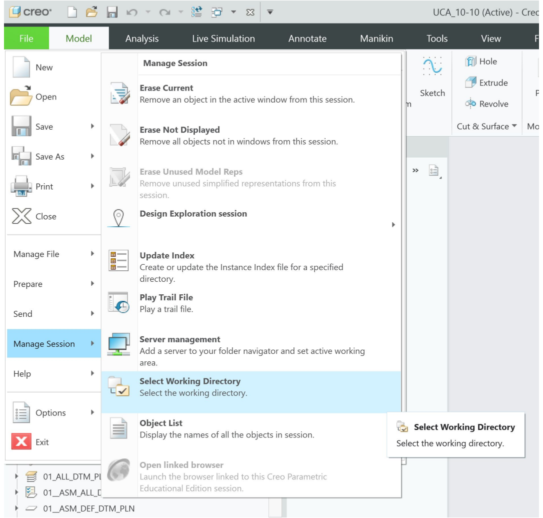
1. Click Select Working Directory → navigate to your spring_ws_final folder you unzipped → OK
2. Click File Open → STEERING_ASSEMBLY_AND_POINTS.asm
   a. Notice all the assembly parts are also in the main spring_ws_final folder!

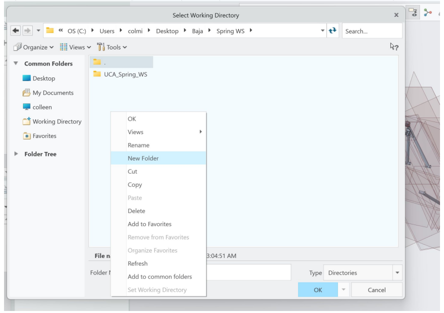


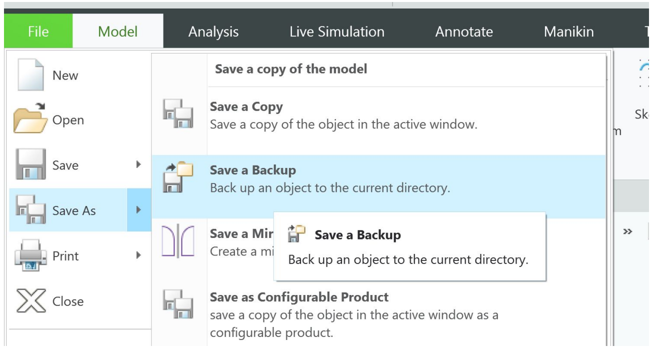


Working Directories

1. File → Manage Session → select working directory → backtrack out of UCA Spring Ws and right click → new folder → name whatever you'd like! → OK
2. File → save as → save a Backup 😼

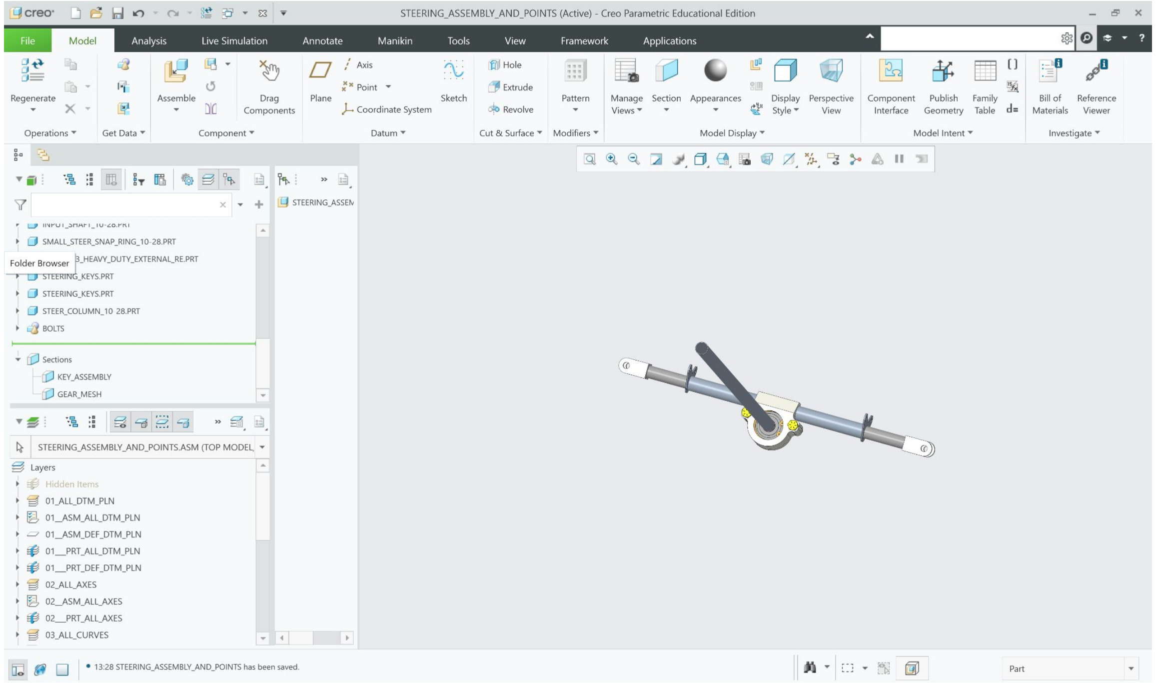




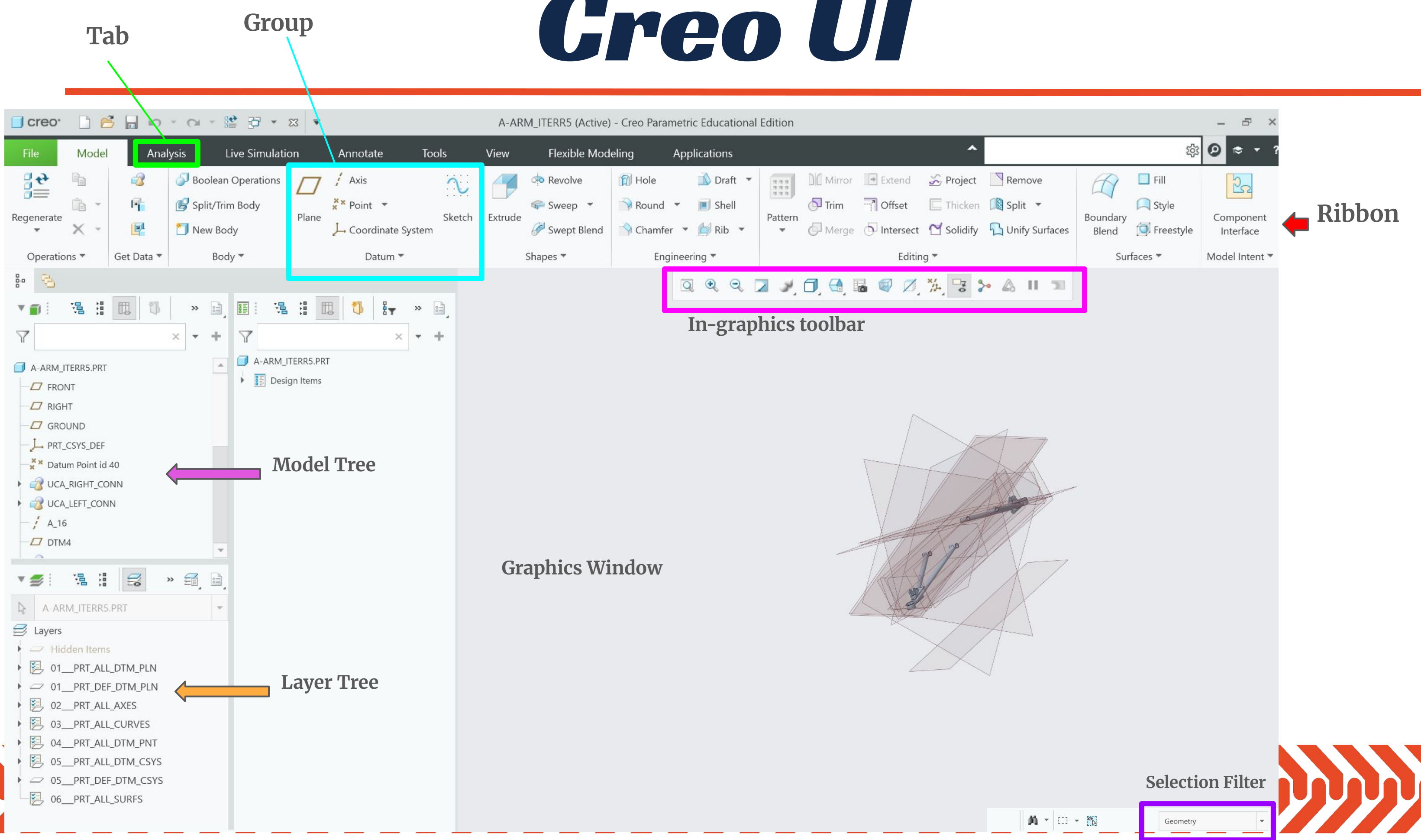


* make sure to file → manage session → erase not displayed

WTF am I looking at



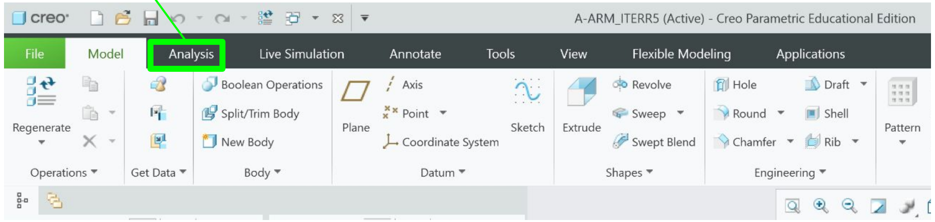
Creo UI



Tab
Group
A-ARM_ITERR5 (Active) - Creo Parametric Educational Edition
File
Model
Analysis
Live Simulation
Annotate
Tools
View
Flexible Modeling
Applications
Regenerate
Boolean Operations
Split/Trim Body
New Body
Plane
Axis
Point
Coordinate System
Sketch
Extrude
Revolve
Sweep
Swept Blend
Hole
Round
Chamfer
Draft
Shell
Rib
Pattern
Mirror
Trim
Merge
Extend
Offset
Intersect
Project
Thicken
Solidify
Remove
Split
Unify Surfaces
Boundary Blend
Fill
Style
Freestyle
Component Interface
Operations
Get Data
Body
Datum
Shapes
Engineering
Editing
Surfaces
Model Intent
Ribbon
In-graphics toolbar
Model Tree
A-ARM_ITERR5.PRT
FRONT
RIGHT
GROUND
PRT_CSYS_DEF
Datum Point id 40
UCA_RIGHT_CONN
UCA_LEFT_CONN
A_16
DTM4
Design Items
Graphics Window
Layers
Hidden Items
01__PRT_ALL_DTM_PLN
01__PRT_DEF_DTM_PLN
02__PRT_ALL_AXES
03__PRT_ALL_CURVES
04__PRT_ALL_DTM_PNT
05__PRT_ALL_DTM_CSYS
05__PRT_DEF_DTM_CSYS
06__PRT_ALL_SURFS
Layer Tree
Selection Filter
Geometry

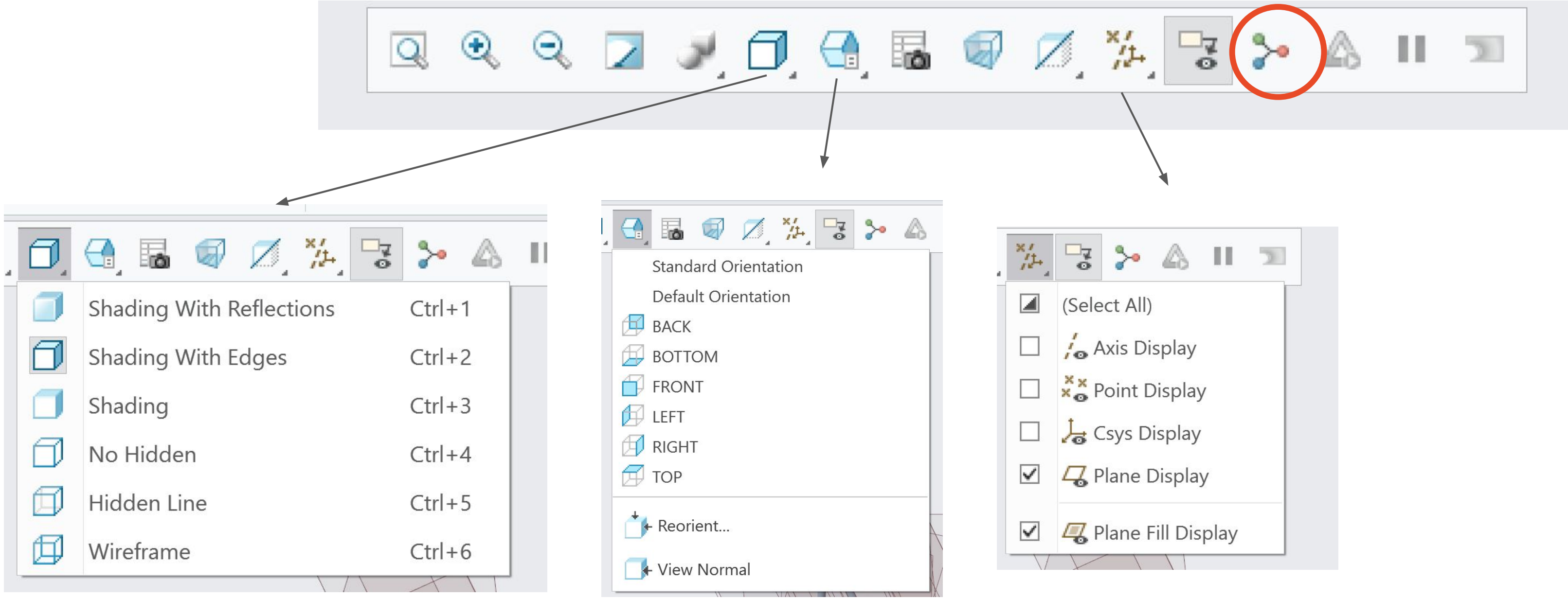
Creo UI – Tab

Tab



- Home base of what you're accessing
- **Model** has most things
  - sketching , extruding, etc
- **Annotate** is clutch for dimensions/GD&T, manufacturing drawings etc
- **Analysis** is clutch for measure tool
- **View** is nice for making things pretty colors and making section views

Creo UI – In-graphics toolbar

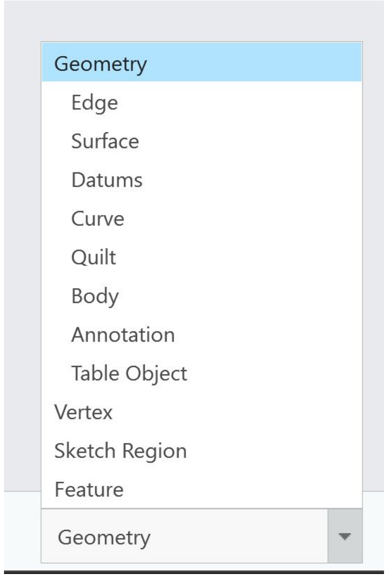


Display style ^

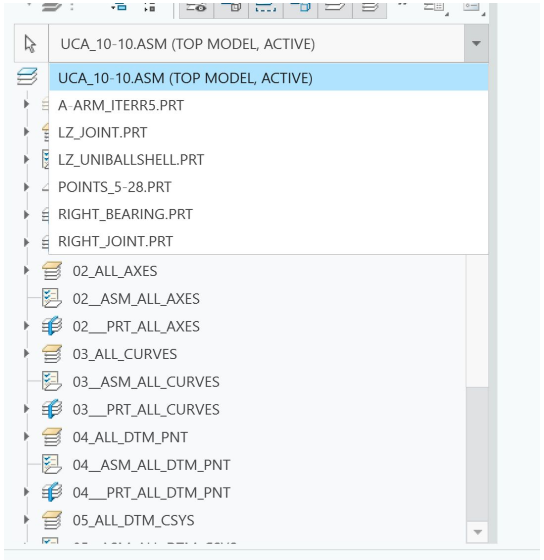
Saved Orientation ^^

Datum Display Styles

Creo UI – Selection filter & Layer tree

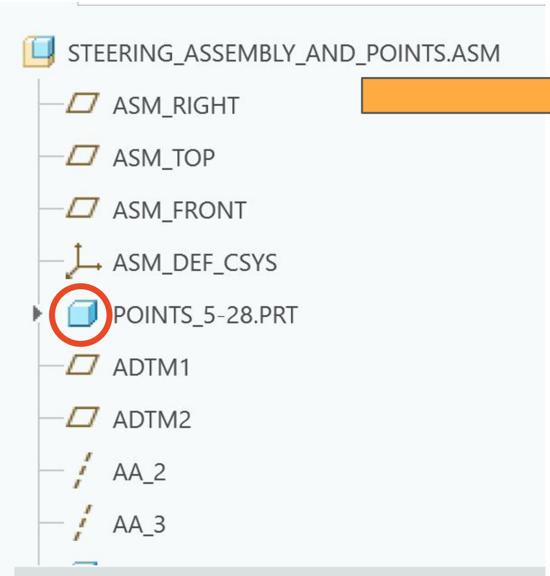


Selection Filter

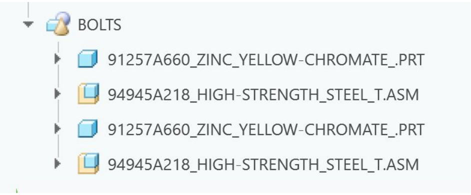


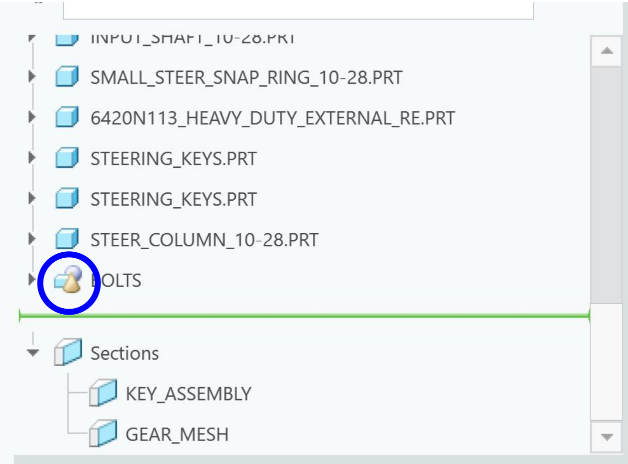
Layer Tree

Creo UI – Model Tree



Datum Planes, Coordinate System



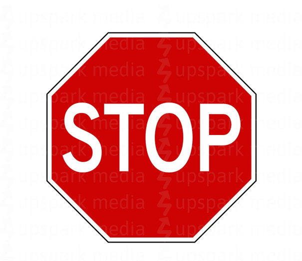


Parts in an assembly ^^

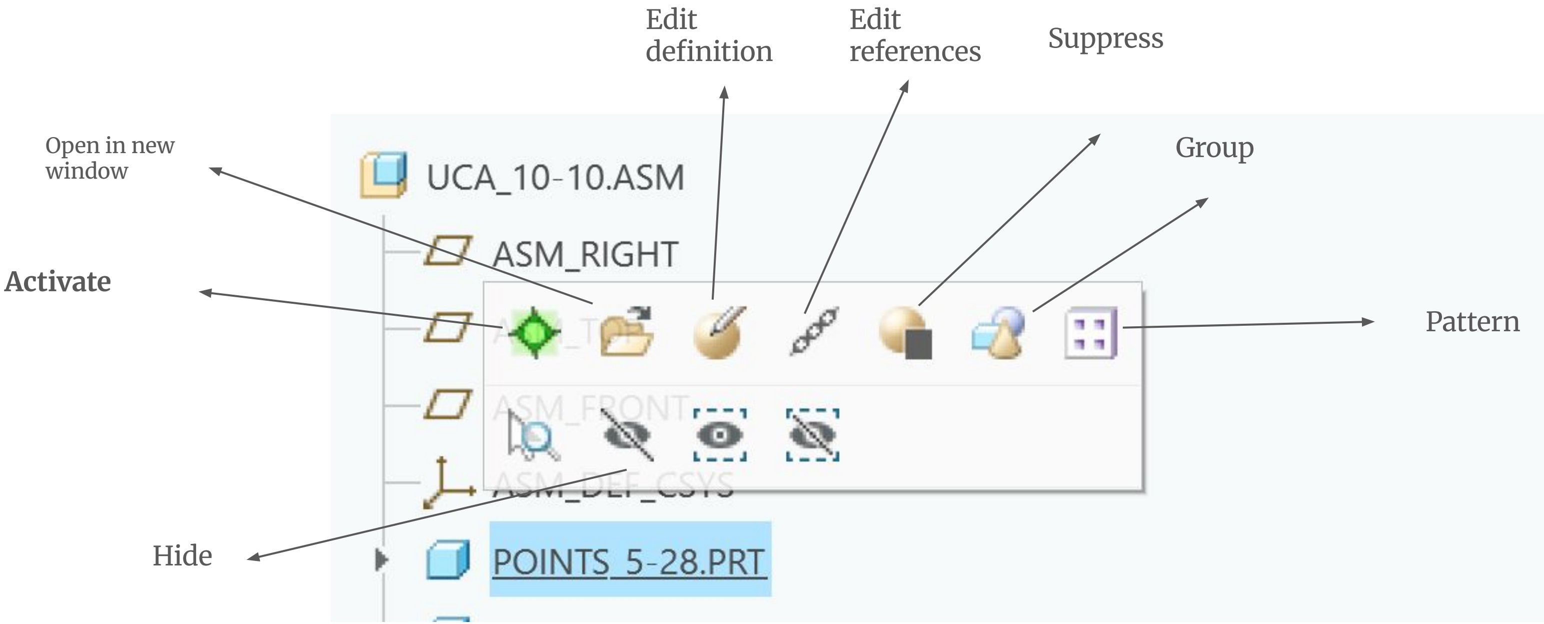
Ts chronological!

Creo UI – Mouse/General functions

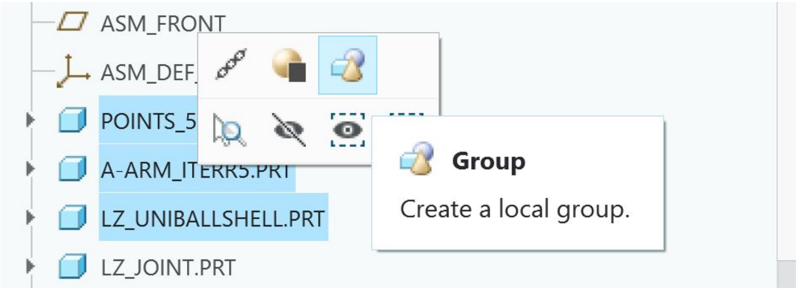
- Scroll wheel - **zoom in and out**
- Middle click - **rotate part** (around where you clicked/center of page)
- Shift & middle click - **translate up/down**
- Click something, hold shift and click a component way further down → selects/deselects all of them
  - Ctrl+click selects one/deselects one item
- Ctrl+click selects multiple things

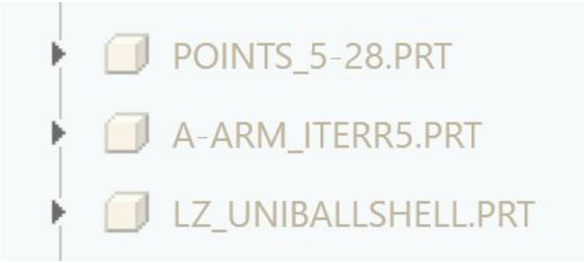


Assembly Navigation!!!!

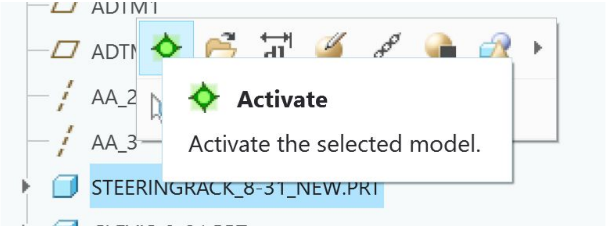


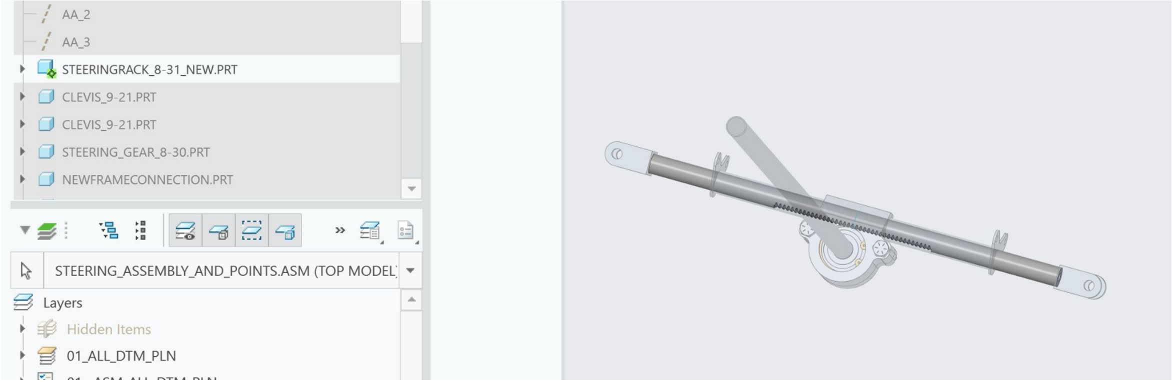
Assy Navigation – Group, Hide, Pattern





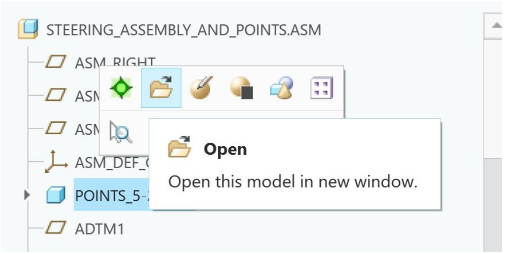
Assy Navigation – Activate

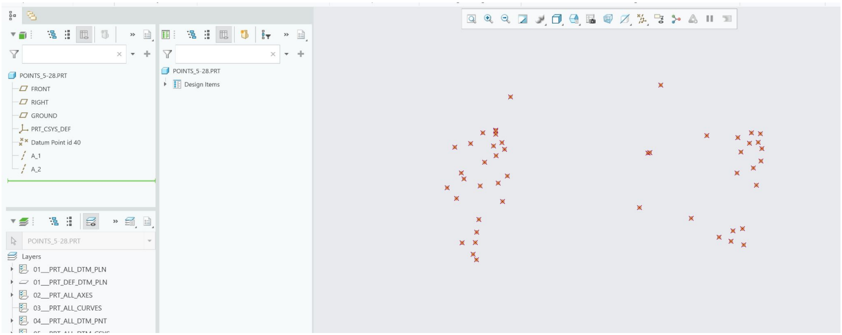




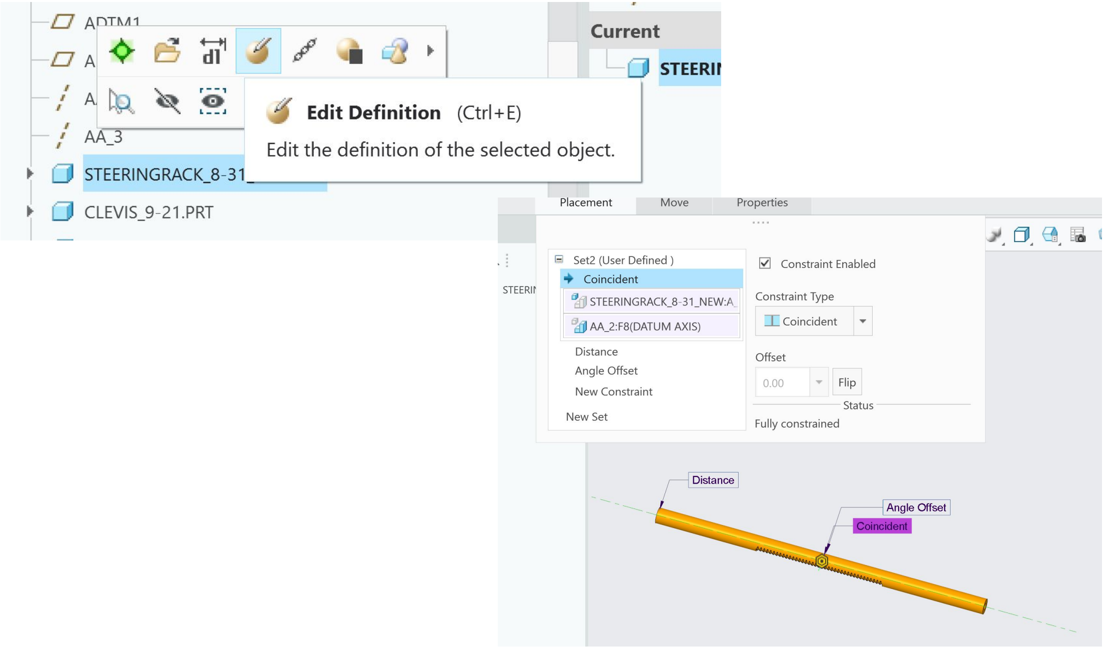
- TS crucial
- If you want to edit just one part in an assembly, use this!
- Only has internal/previous references
- Looks like ts
- Don’t fret if everything blows up after you change one part

Assy Navigation – Open in New Window



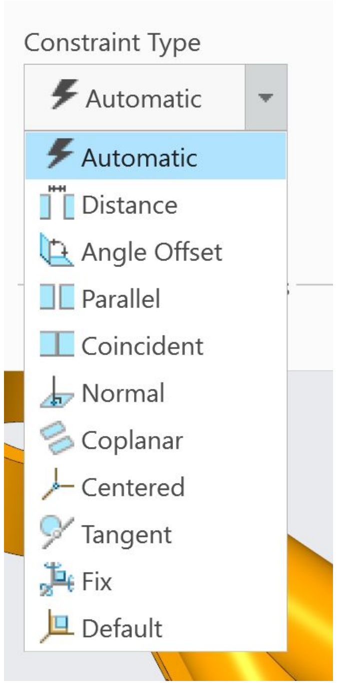


Assy Navigation – Edit Definition



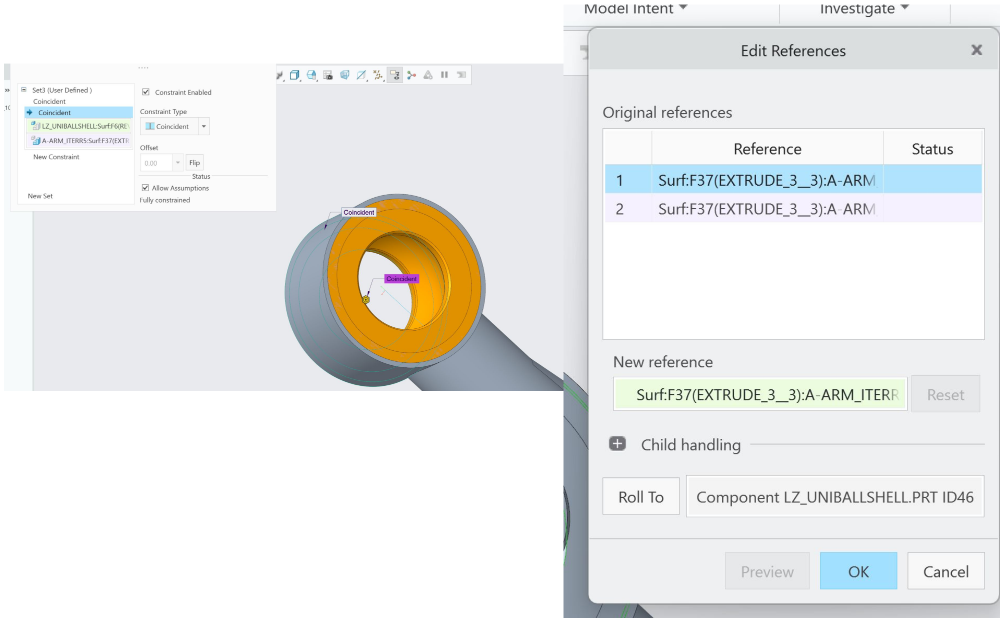
- Multiple meaning word
- In assemblies, revolves around constraints!
    - Constraints?

Brief aside… constraints



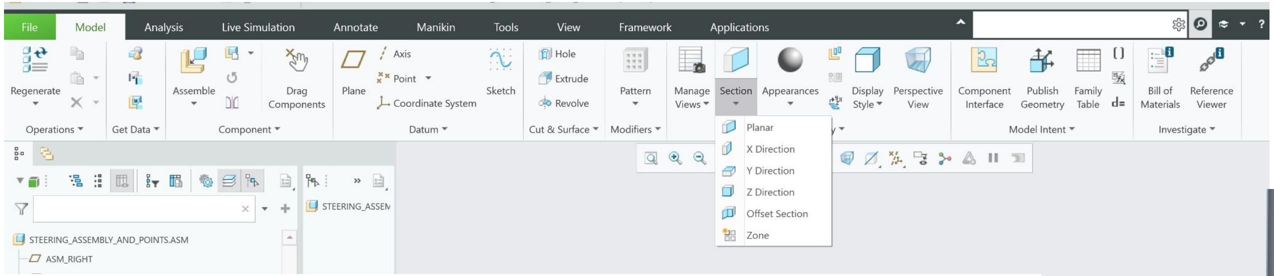
- How assemblies assemble!!
- Have to jump through these hoops every time you want two parts to fit in relation to each other
- Don't worry if they all blow up! Thats what **edit references** is for…………

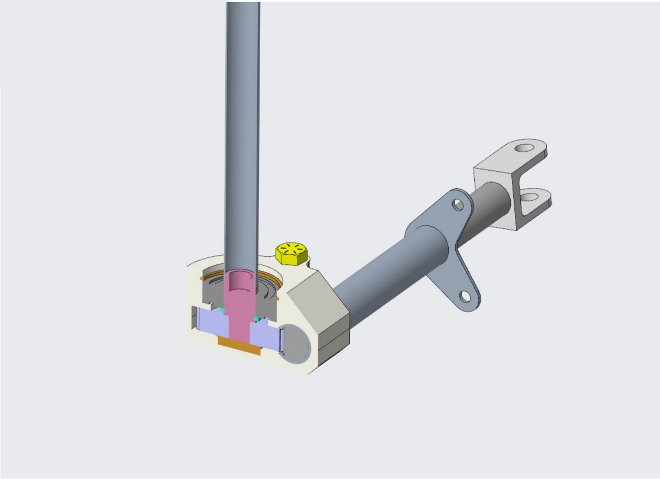
Assy Navigation – Edit References



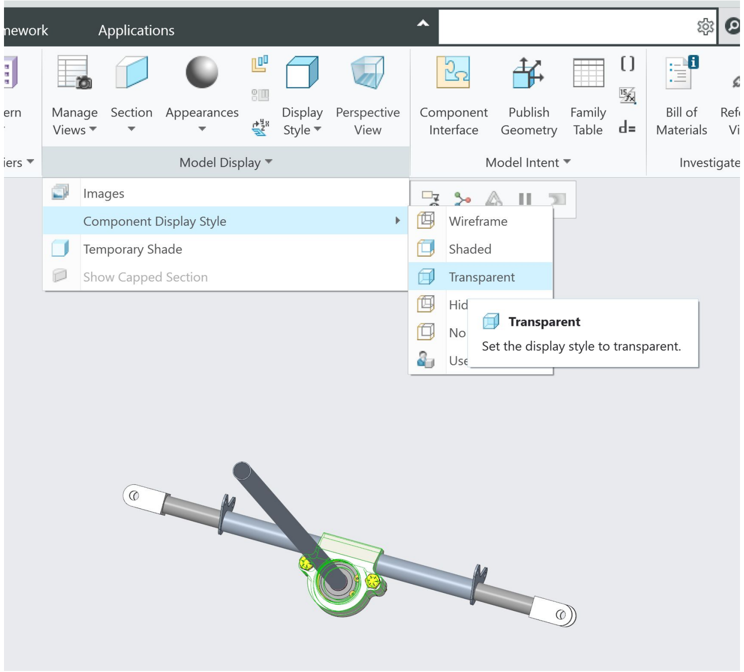
- When updating parts in an assembly, if dims change, constraints become invalid and BLOW UP
- Edit references can quickly replace what's blown up with what's new and can be done on a major scale!

Gen. Navigation → Section views





How to make pieces transparent



More Baja related...

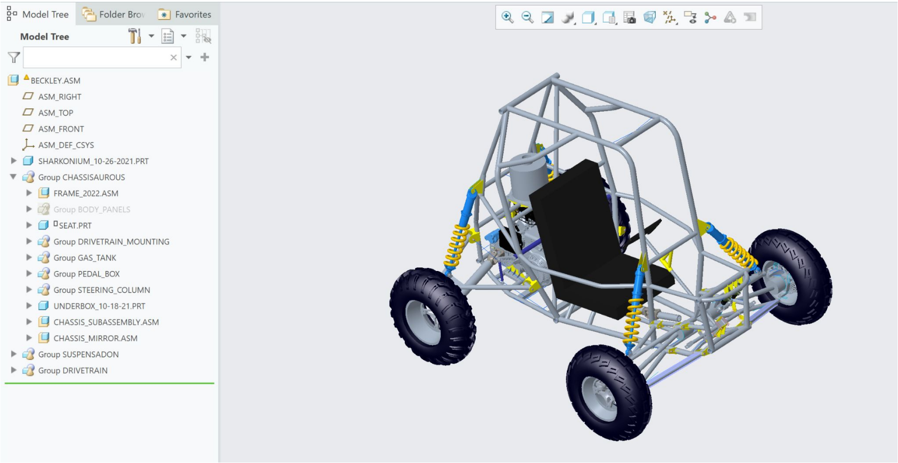
Hi you'll be modeling things quite shortly after this stay with me now

General CAD tips

- The order you make things MATTERS. A LOT.
- Reference things to **planes**! Make dimensions easily editable!
    - It's way better to constrain lots of things in the beginning in order to be able to edit one dimension later on
- Think about manufacturing / assembling
    - If it's a beautiful part but we can't make it or get it in the car, it's no good
- Label things in the model tree clearly
    - Helps you / your sub-team lead modify things later

Chain of command

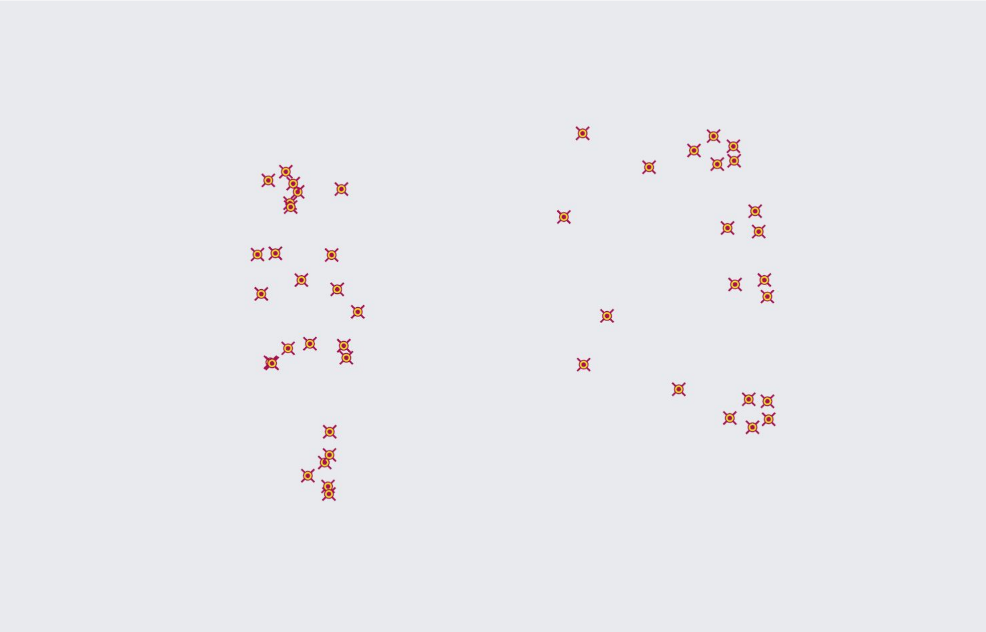
- We have a google folder in the drive which sub-team leads and senior designers check in and out
- Individual part designers send their parts to their sub-team leads who update them in the model

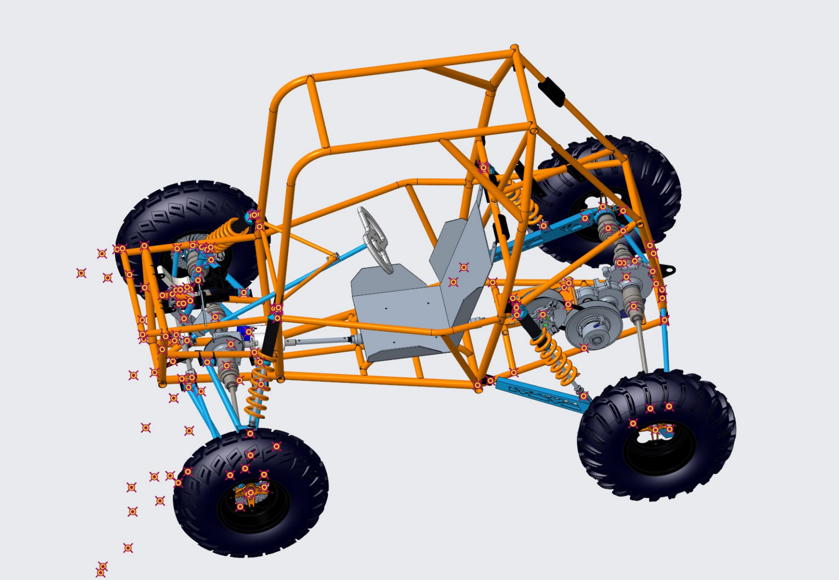


Design to points

- Certain components get made to fit to defined points
    - Suspension points - you can see these in the assembly!
        - Points 5-28
    - Chassis points
- Each point refers to something in the final model:
    - Start, end, or bend of a tube
    - Rotation axis of an A-arm, where a bolt goes
    - Connection point of a rod, shock, etc.
- Designs should be tailored to the points, and flexible to changes
- Most things are symmetrical so one side is designed and then mirrored

Points and Curves





Importing Points

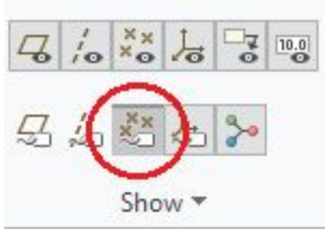
- Points designers send out a part file with all the datums defined
- **If you're starting a new part**, just open the file and start modeling
- Save as a copy!

This is not yet relevant for most of you

<ins>**Important:**</ins>

- Make sure you know which points matter for your part!
- If you don't know exactly what the point refers to in real life, ASK!

In the view tab, click this to see the point names



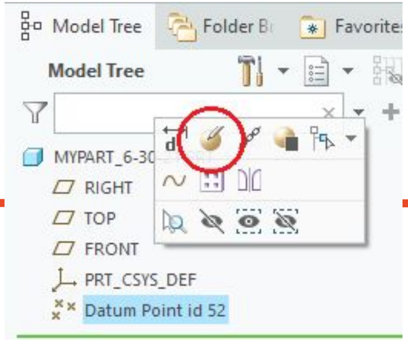
Updating Points

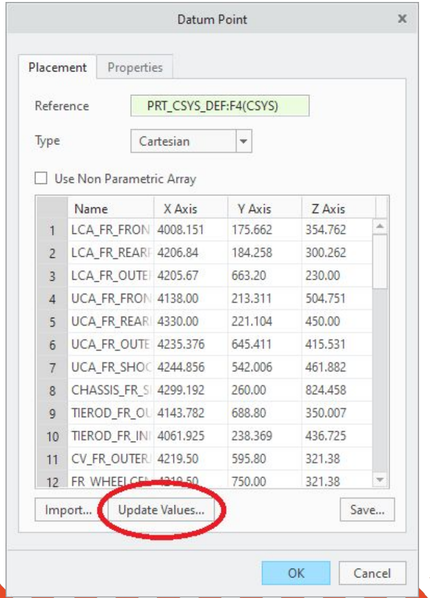
When a new points file is sent out:

- In the new points model, edit the points and “update values”, then copy all the numbers in the text file
- Open your component model and go to the same text file, then paste the coordinates you copied

You may then have to make changes to your part; depends on how flexible your design is

Hi the same thing applies to this AND if you’re curious there’s an entire points slideshow linked in the references!





Modeling Walkthrough!

Brief poll...

Hi I have a CAD project that I'd like to work on now with guidance from everyone

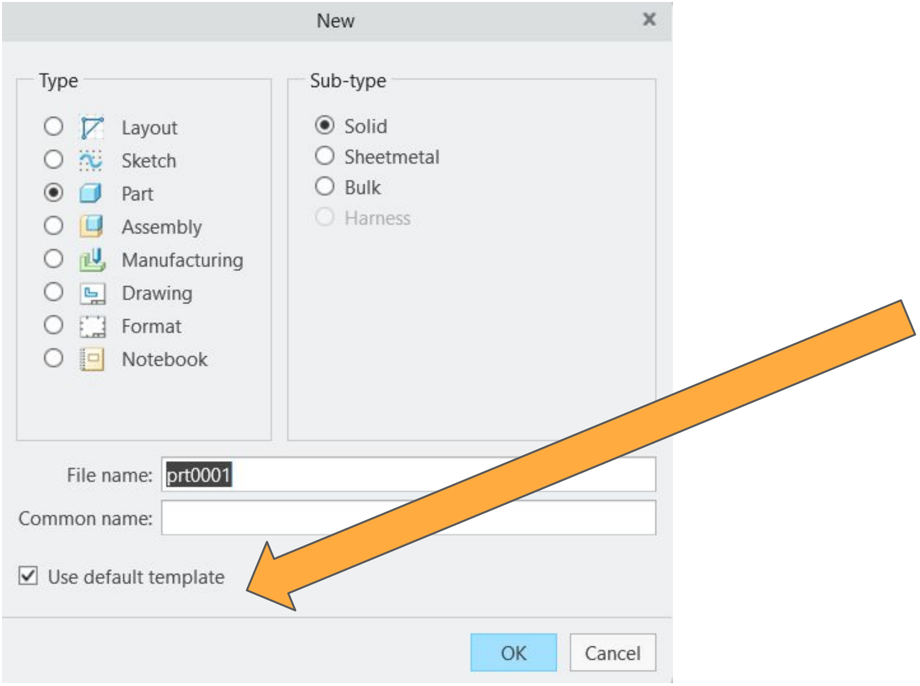
vs.

I want to do a walkthrough with everyone and mess around with this steering part!

Modelling

- **Purpose**
  - Assembly
  - Requirements
    - Features + Dimensions
- **Start big and hone in**
  - Construction Lines
  - Reference Points (Projections from other parts)

Modelling



Default Template:

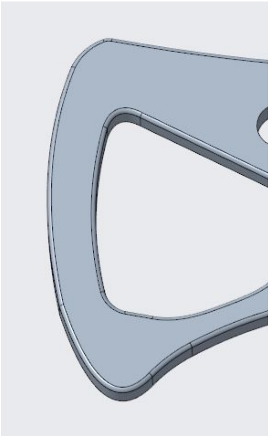
inlbs_part_solid

Modelling – Steering Wheel

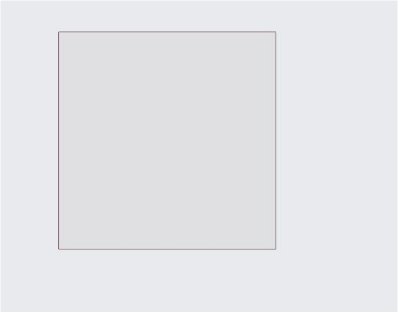
Grip cavities

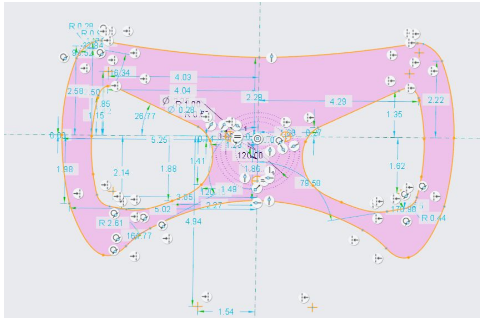
Button Holes

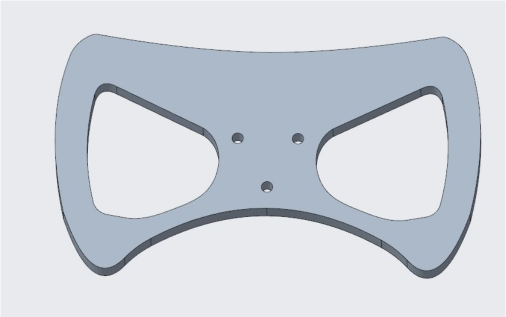
Quick Release Mount

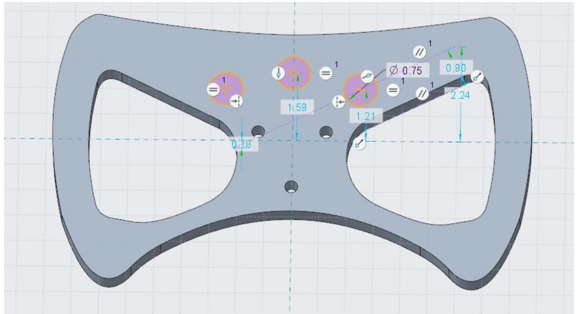


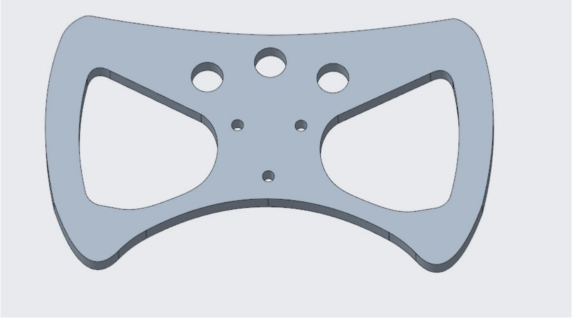
Steps

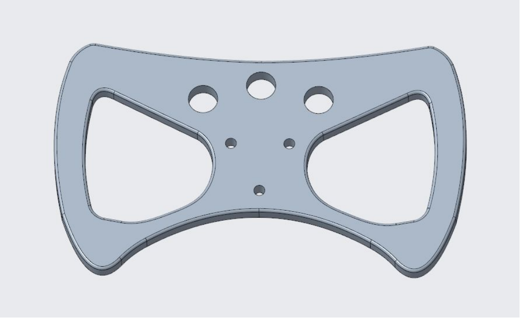




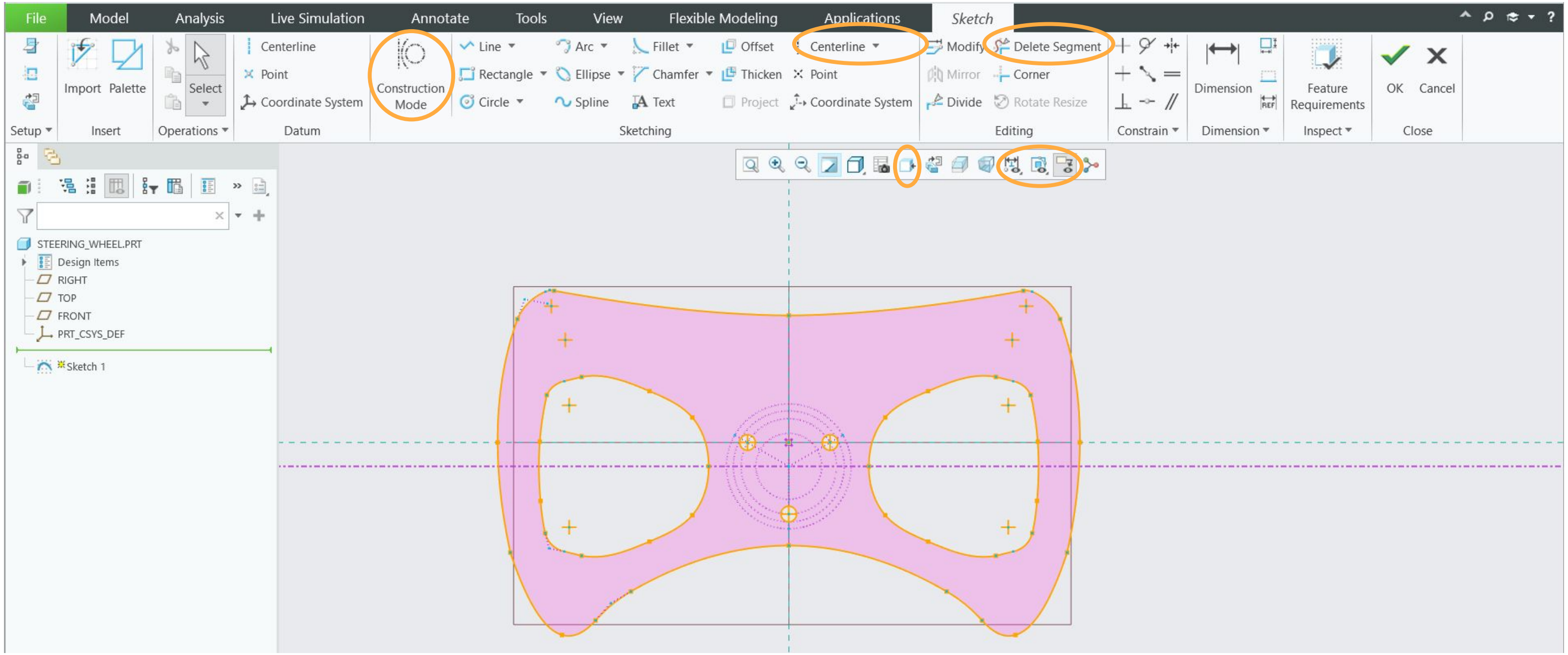








Modelling



File
Model
Analysis
Live Simulation
Annotate
Tools
View
Flexible Modeling
Applications
Sketch
Import
Palette
Select
Centerline
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Coordinate System
Construction Mode
Line
Rectangle
Circle
Arc
Ellipse
Spline
Fillet
Chamfer
Text
Offset
Thicken
Project
Centerline
Point
Coordinate System
Modify
Delete Segment
Mirror
Corner
Divide
Rotate Resize
Dimension
Feature Requirements
OK
Cancel
Setup
Insert
Operations
Datum
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Editing
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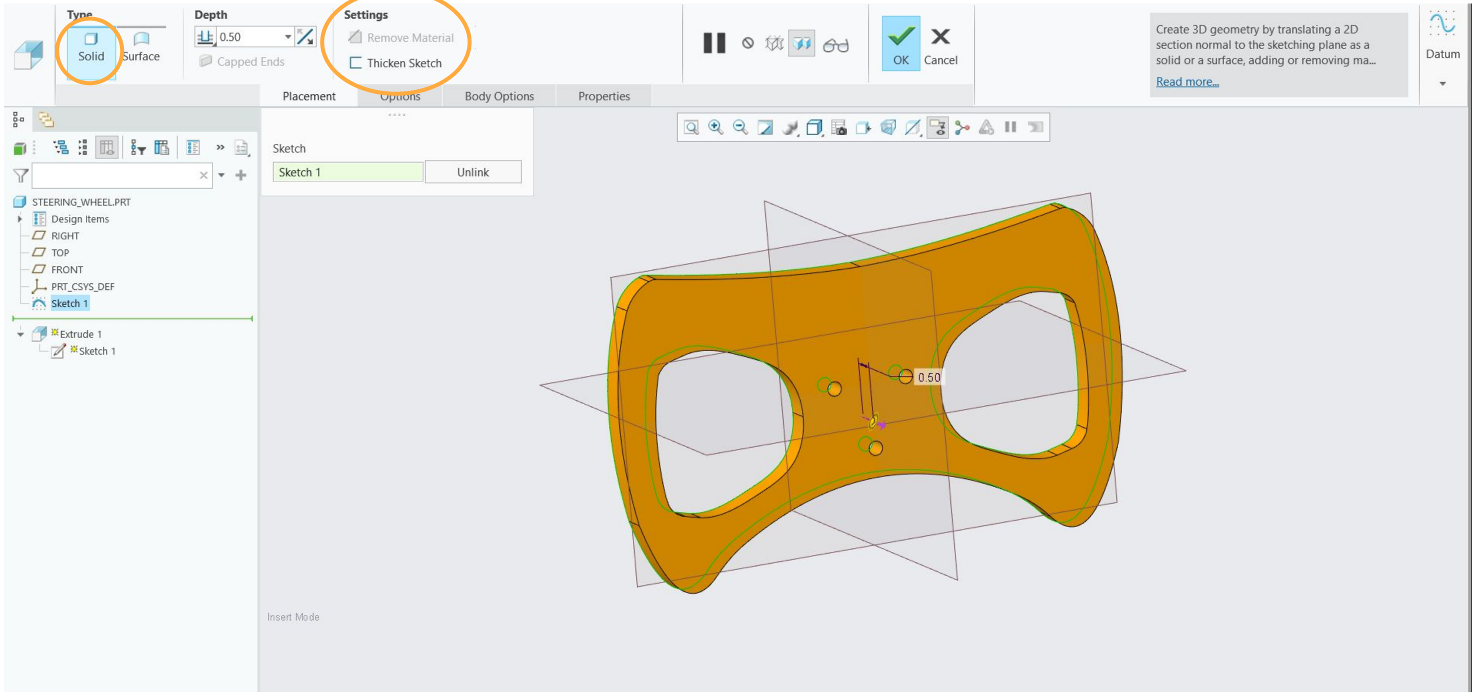
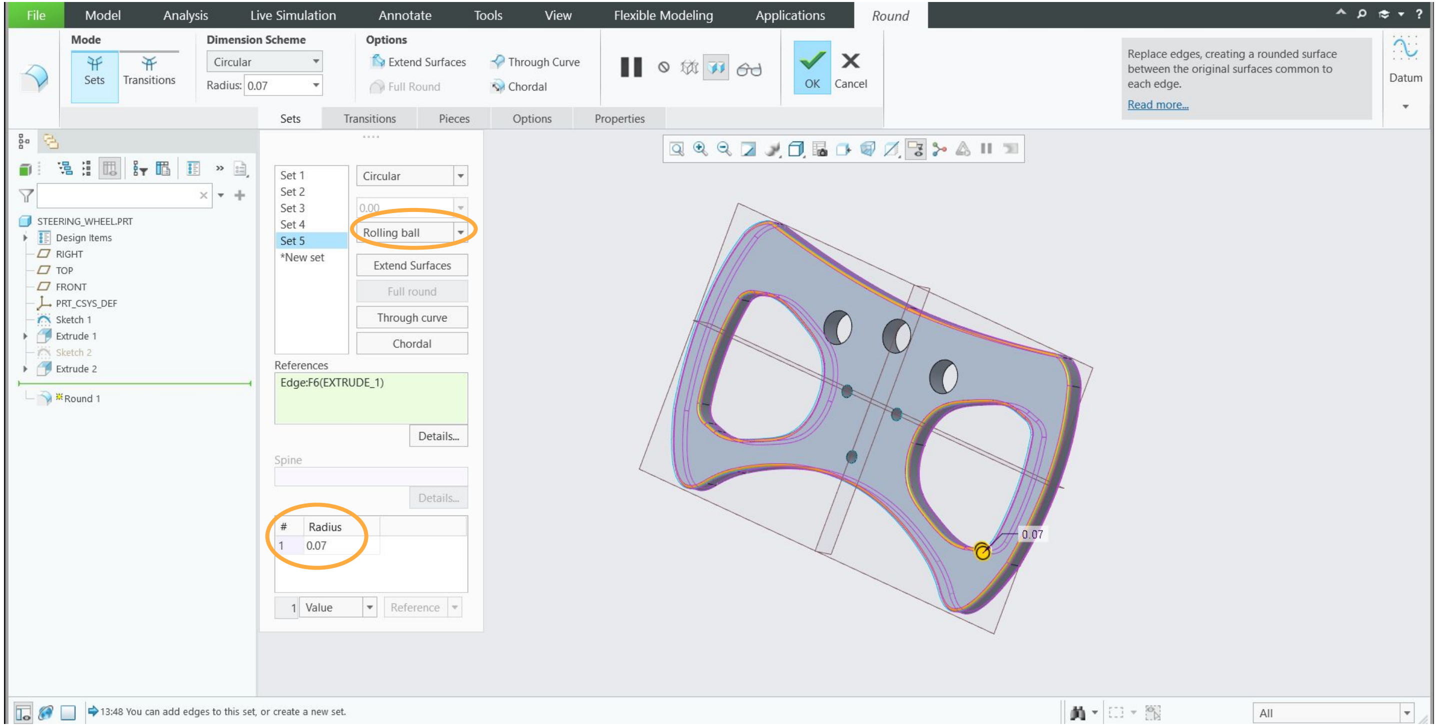
Close
STEERING_WHEEL.PRT
Design Items
RIGHT
TOP
FRONT
PRT_CSYS_DEF
Sketch 1

Modelling



Type
Solid
Surface
Depth
0.50
Capped Ends
Settings
Remove Material
Thicken Sketch
OK
Cancel
Create 3D geometry by translating a 2D section normal to the sketching plane as a solid or a surface, adding or removing ma...
Read more...
Datum
Placement
Options
Body Options
Properties
Sketch
Sketch 1
Unlink
STEERING_WHEEL.PRT
Design Items
RIGHT
TOP
FRONT
PRT_CSYS_DEF
Sketch 1
Extrude 1
Sketch 1
0.50
Insert Mode

Modelling


File
Model
Analysis
Live Simulation
Annotate
Tools
View
Flexible Modeling
Applications
Round
Mode
Sets
Transitions
Dimension Scheme
Circular
Radius: 0.07
Options
Extend Surfaces
Through Curve
Full Round
Chordal
OK
Cancel
Replace edges, creating a rounded surface between the original surfaces common to each edge.
Read more...
Datum
Sets
Transitions
Pieces
Options
Properties
STEERING_WHEEL.PRT
Design Items
RIGHT
TOP
FRONT
PRT_CSYS_DEF
Sketch 1
Extrude 1
Sketch 2
Extrude 2
Round 1
Set 1
Set 2
Set 3
Set 4
Set 5
*New set
Circular
0.00
Rolling ball
Extend Surfaces
Full round
Through curve
Chordal
References
Edge:F6(EXTRUDE_1)
Details...
Spine
Details...
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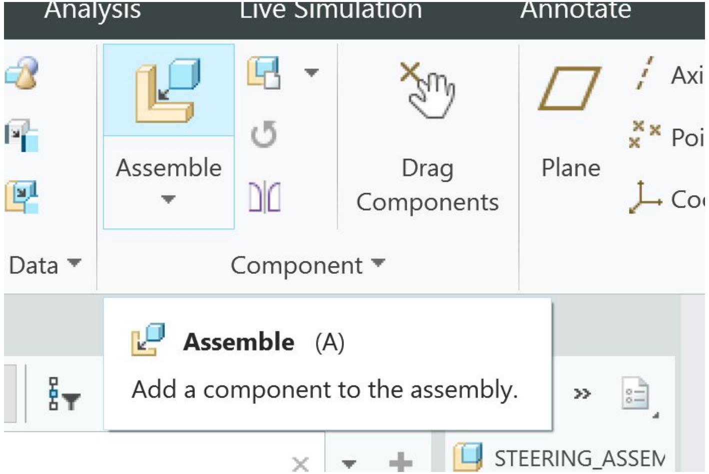
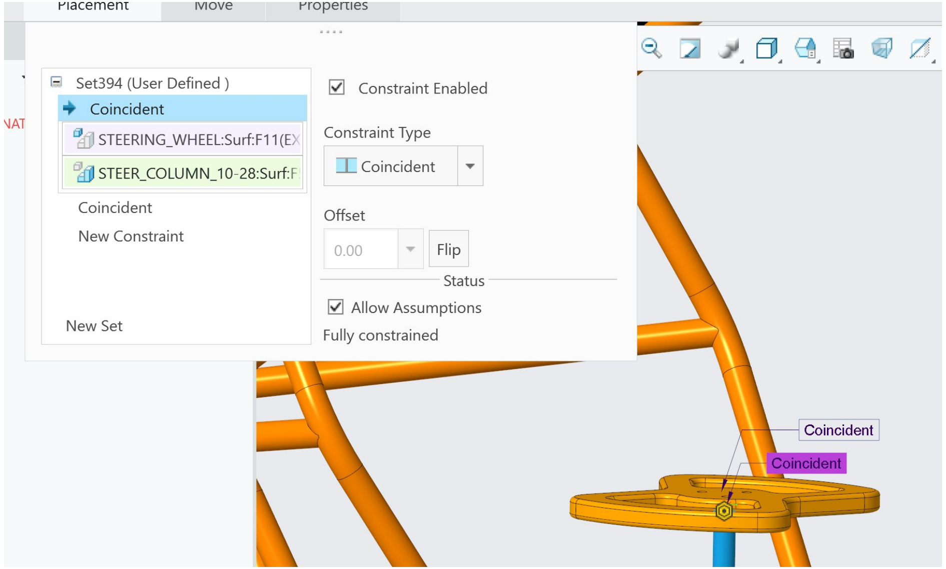
1 Value
Reference
0.07
13:48 You can add edges to this set, or create a new set.
All

Constrain to assy





Troubleshooting

- **Esc key** is your best friend
- **Erase not displayed**!! If you want to edit a part and you've pulled it up recently
  - "Creo retains a model in temporary memory until the model is erased, or until the application is closed."
- **Update references!!!**
  - Can also be evil but is a necessary evil
- **Insert here** is also your best friend
- Pls pls be clean and neat and tidy
- **Lock dimensions** if you want them to stay PUT

More Summer Design Info!

Q&A!

Thanks for popping out!!!

- Summer Design Bouta be a Movie

Fill out Summer Design Interest Form!

Due Monday @4pm